



THE UNIVERSITY OF ARIZONA
COLLEGE OF SCIENCE

Theoretical Astrophysics
Program

2027

TAP Travel / Training Grants

APPLICATIONS NOW
BEING ACCEPTED

Deadline to apply: March 9 by 5pm MST

Available for use

July 1, 2026 - May 31, 2027

(travel/training may occur prior to July 1)



APPLY NOW

Notes on Proposal Topic Selection

Can be your own current research project

Can be an idea already published in paper if that topic is related to your thesis (distinct from Assign 2, 3)

Can be theory —> map to JWST/HST science themes but can reference contributions of other telescopes.

Can be observations for a different observatory (discuss with me)

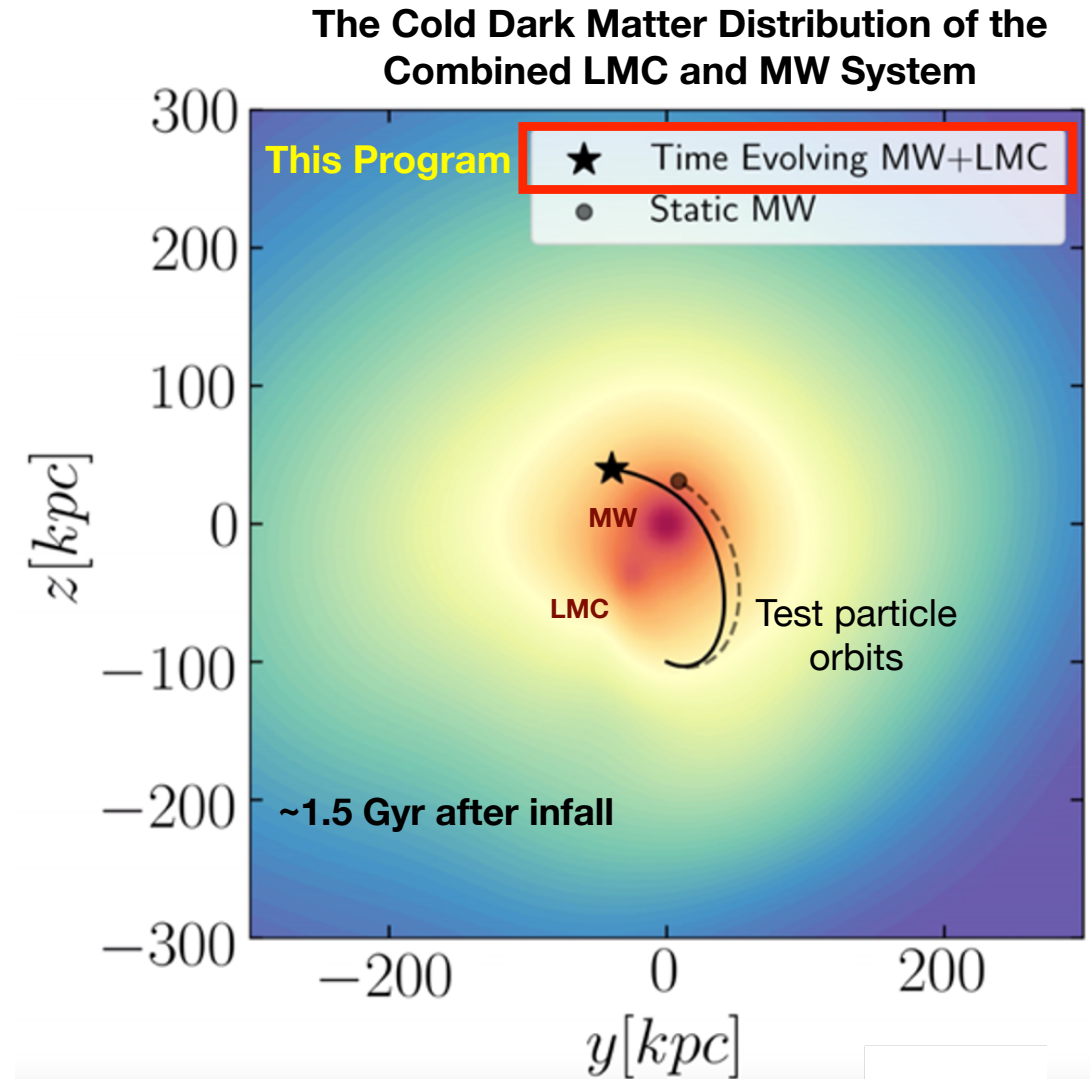
Assignment 4: Example

Assign 4 Part A. Figure Caption format

- First sentence: Bold faced takeaway that highlights the broader impact of the program.
- Second sentence: Description of what is plotted
- Next: Statement of the problem related to the target/plot
- Last sentence(s) is(are) the Goal (“This program will ...” or “We propose to”) AND the Importance of the solution. Underline the Goal statement.

Figure & Caption

The motions of all objects in orbit about the **Milky Way (MW)** are impacted by the dark matter distortions caused by the recent infall of the **LMC**. The color scale illustrates the combined dark matter distribution of the MW and the LMC, 1.5 Gyr after the LMC's infall to the MW. The dashed line illustrates the orbit of a test particle, assuming a static MW. The solid line illustrates the orbit of the same test particle, but now accounting for the deformation of the MW's dark matter caused by the LMC. Ignoring the time evolving dark matter perturbations from the LMC will cause errors in orbital calculations. This program will build high-resolution N-body simulations to quantify the time evolving MW+LMC dark matter potential. Combined with HST/Gaia astrometry, the proposed new potentials will enable the orbit reconstruction of *all* MW halo tracers (globular clusters, satellites, streams) necessary to constrain the dark matter distribution of our Galaxy.



NOTE: there should have been a color bar

Figure & Caption

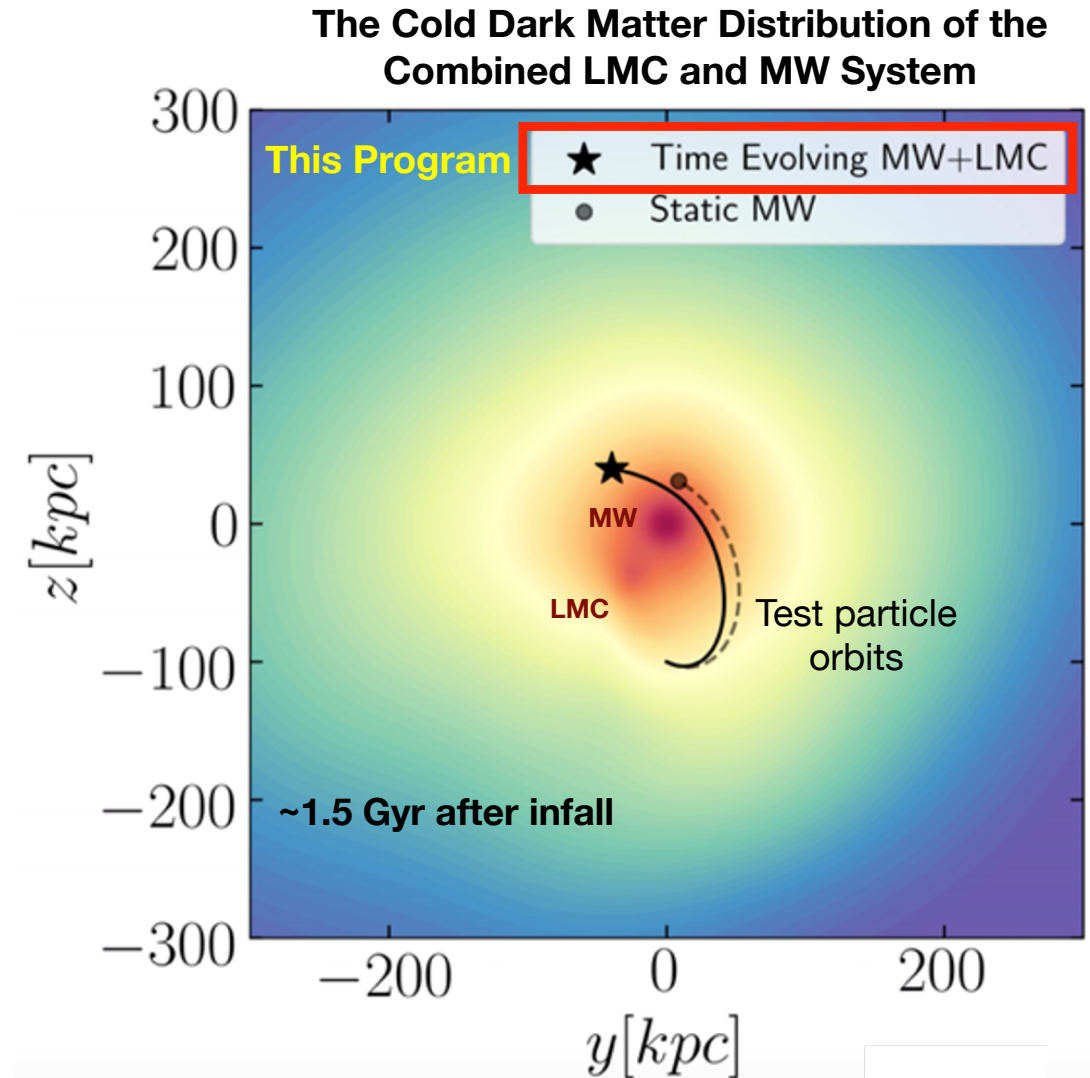
First Sentence, Takeaway Point : The motions of all objects in orbit about the Milky Way (MW) are impacted by the dark matter distortions caused by the recent infall of the LMC.

Description of what is plotted: The color scale illustrates the combined dark matter distribution of the MW and the LMC, 1.5 Gyr after the LMC's infall to the MW. The dashed line illustrates the orbit of a test particle, assuming a static MW. The solid line illustrates the orbit of the same test particle, but now accounting for the deformation of the MW's dark matter caused by the LMC.

Problem: Ignoring the time evolving dark matter perturbations from the LMC will cause errors in orbital calculations.

Goal: This program will build high-resolution N-body simulations to quantify the time evolving MW+LMC dark matter potential.

Importance: Combined with HST/Gaia astrometry, the proposed new potentials will enable the orbit reconstruction of *all* MW halo tracers (globular clusters, satellites, streams) necessary to constrain the dark matter distribution of our Galaxy.



NOTE: there should have been a color bar

Talk Abstract

Constraining Dark Matter in the Era of Precision Astrometry: New Models for the Milky Way

The Hubble Space Telescope (HST) and Gaia have ushered a new era of high precision astrometry, providing accurate 3D velocity measurements of objects orbiting within the Milky Way's dark matter halo. When combined with detailed models of the MW's dark matter potential, such measurements become high-precision tools to constrain the dark matter mass profile of the MW. However, the MW hosts a massive satellite galaxy, the LMC, that changes the shape of the MW's dark matter potential in a non-symmetrical, time evolving manner. In this talk, I will demonstrate that high resolution simulations can be used to quantify the time evolving structure of the MW's dark matter potential owing to the influence of the LMC, enabling the orbit reconstruction of halo tracers that accurately accounts for the LMC's influence for the first time. The era of high-precision astrometry has arrived, yet we do not currently have an appropriate theoretical framework to study the assembly history of MW-like galaxies in the presence of massive satellite perturbers. The results of this work are critical to ongoing HST/JWST programs of the MW and broader efforts to understand the structure and evolution of the dark matter halo of other systems that host massive satellites.

Talk Abstract

Constraining Dark Matter in the Era of Precision Astrometry: New Models for the Milky Way

FACT:

IMPORTANCE OF FACT:

PROBLEM:

GOAL:

URGENCY/IMPORTANCE TO SUBFIELD

BROADER IMPACT TO FIELD:

Talk Abstract

Constraining Dark Matter in the Era of Precision Astrometry: New Models for the Milky Way

FACT : The Hubble Space Telescope (HST) and Gaia have ushered a new era of high precision astrometry, providing accurate 3D velocity measurements of objects orbiting within the Milky Way's dark matter halo.

IMPORTANCE OF FACT: When combined with detailed models of the MW's dark matter potential, such measurements become high-precision tools to constrain the dark matter mass profile of the MW.

PROBLEM: However, the MW hosts a massive satellite galaxy, the LMC, that changes the shape of the MW's dark matter potential in a non-symmetrical, time evolving manner.

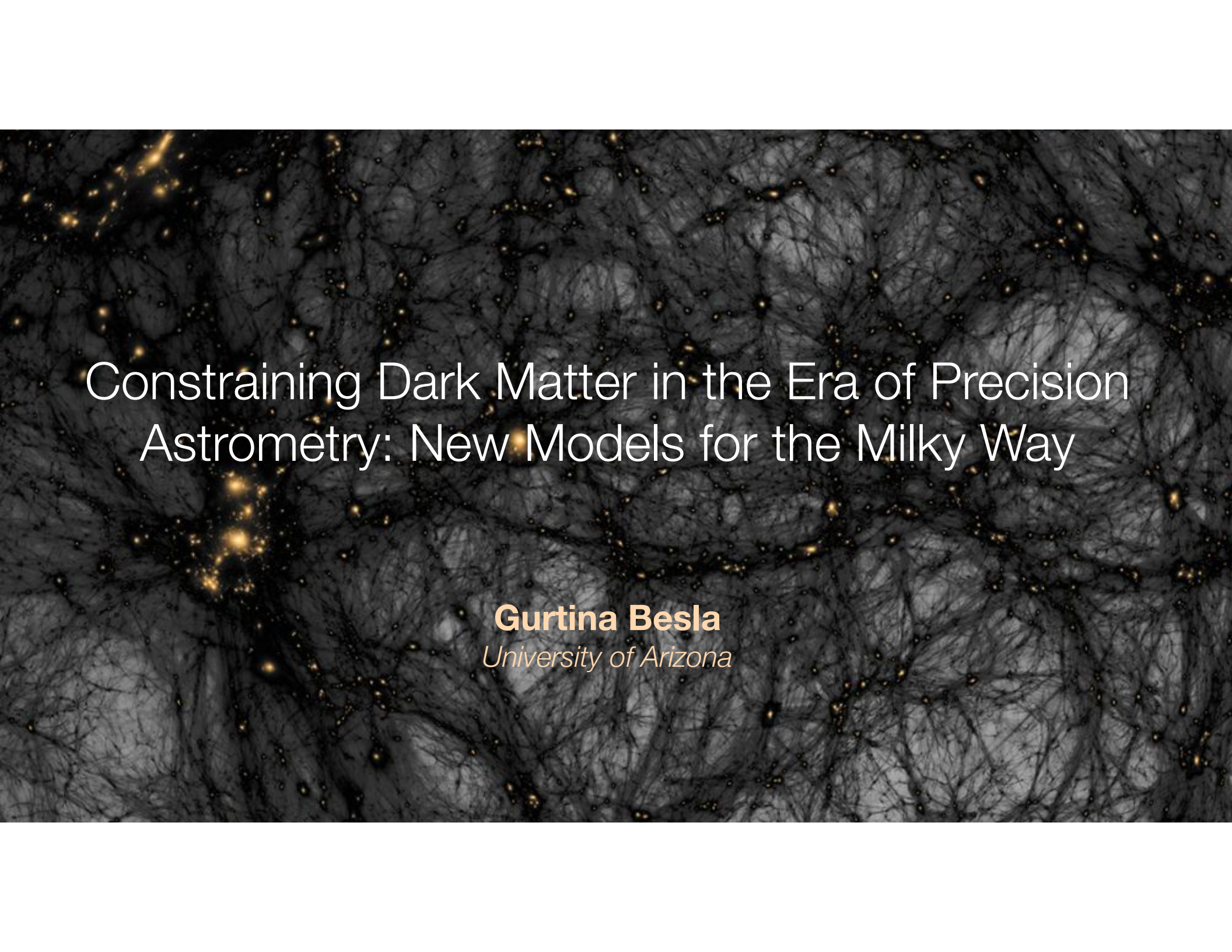
GOAL: In this talk, I will demonstrate that high resolution simulations can be used to quantify the time evolving structure of the MW's dark matter potential owing to the influence of the LMC, enabling the orbit reconstruction of halo tracers that accurately account for the LMC's influence for the first time.

URGENCY/IMPORTANCE TO SUBFIELD: The era of high-precision astrometry has arrived, yet we do not currently have an appropriate theoretical framework to study the assembly history of MW-like galaxies in the presence of massive satellite perturbers.

BROADER IMPACT TO FIELD: The results of this work are critical to ongoing HST/JWST programs of the MW and broader efforts to understand the structure and evolution of the dark matter halo of other systems that host massive satellites.

Framing

- Remember that it is a privilege to have the stage and have the opportunity to share your excitement about your research with an audience.
- Remember that audience members are busy and yet chose to be there to listen to you.

A visualization of the cosmic web, showing a complex network of dark matter filaments and clusters of galaxies. The filaments are represented by a dense, interconnected web of thin, dark lines, while the galaxy clusters are shown as bright, yellowish-orange points of light. The overall background is a dark, textured gray.

Constraining Dark Matter in the Era of Precision Astrometry: New Models for the Milky Way

Gurtina Besla
University of Arizona

A long time ago in a galaxy far,
far away....

The beginning sets the tone.

STAR WARS

Episode IV

A NEW HOPE

*It is a period of civil war.
Rebel spaceships, striking
from a hidden base, have won*

HST and Gaia The Era of Precision Astrometry: *A Moving* Milky Way



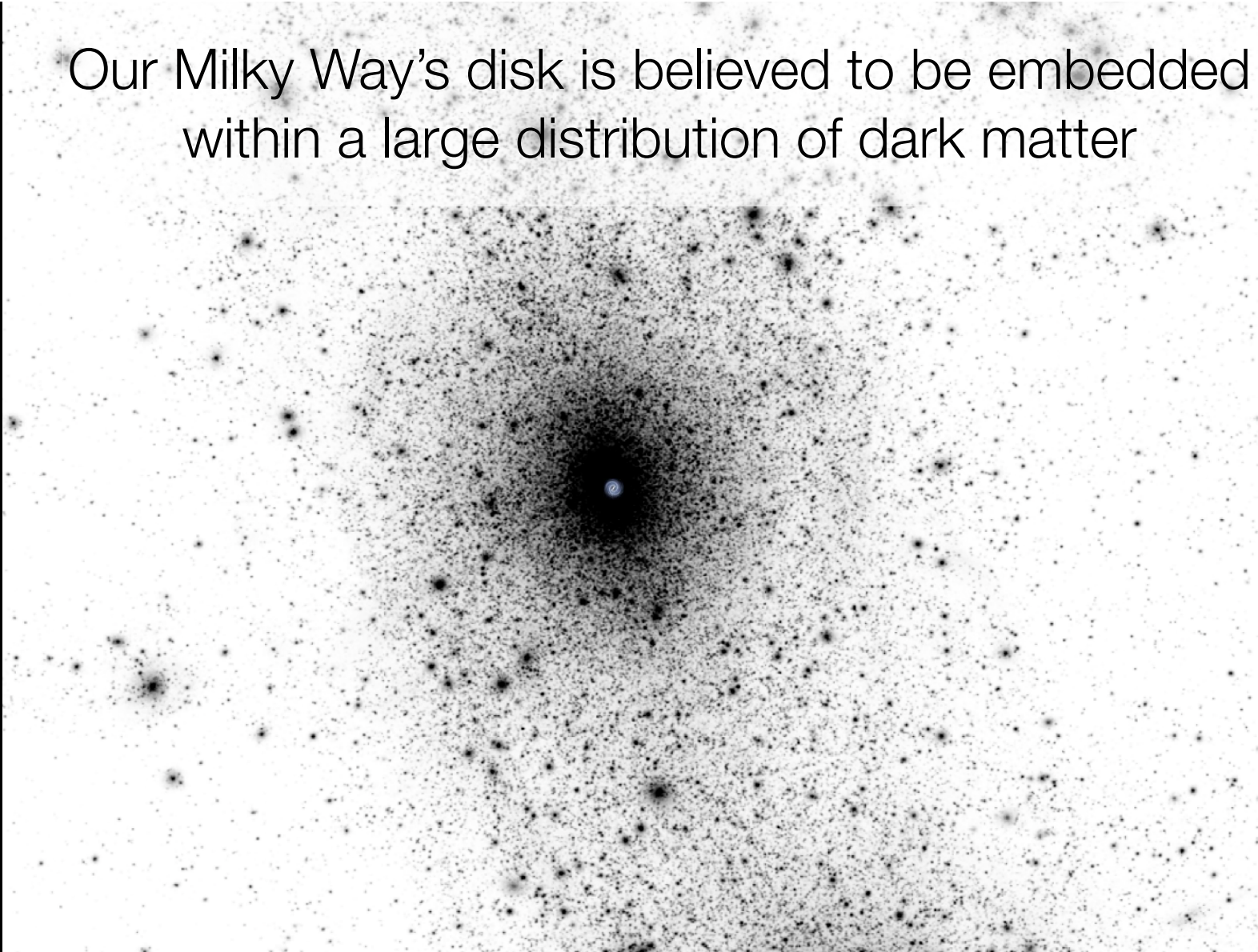
Years from now: 157,500



Fact

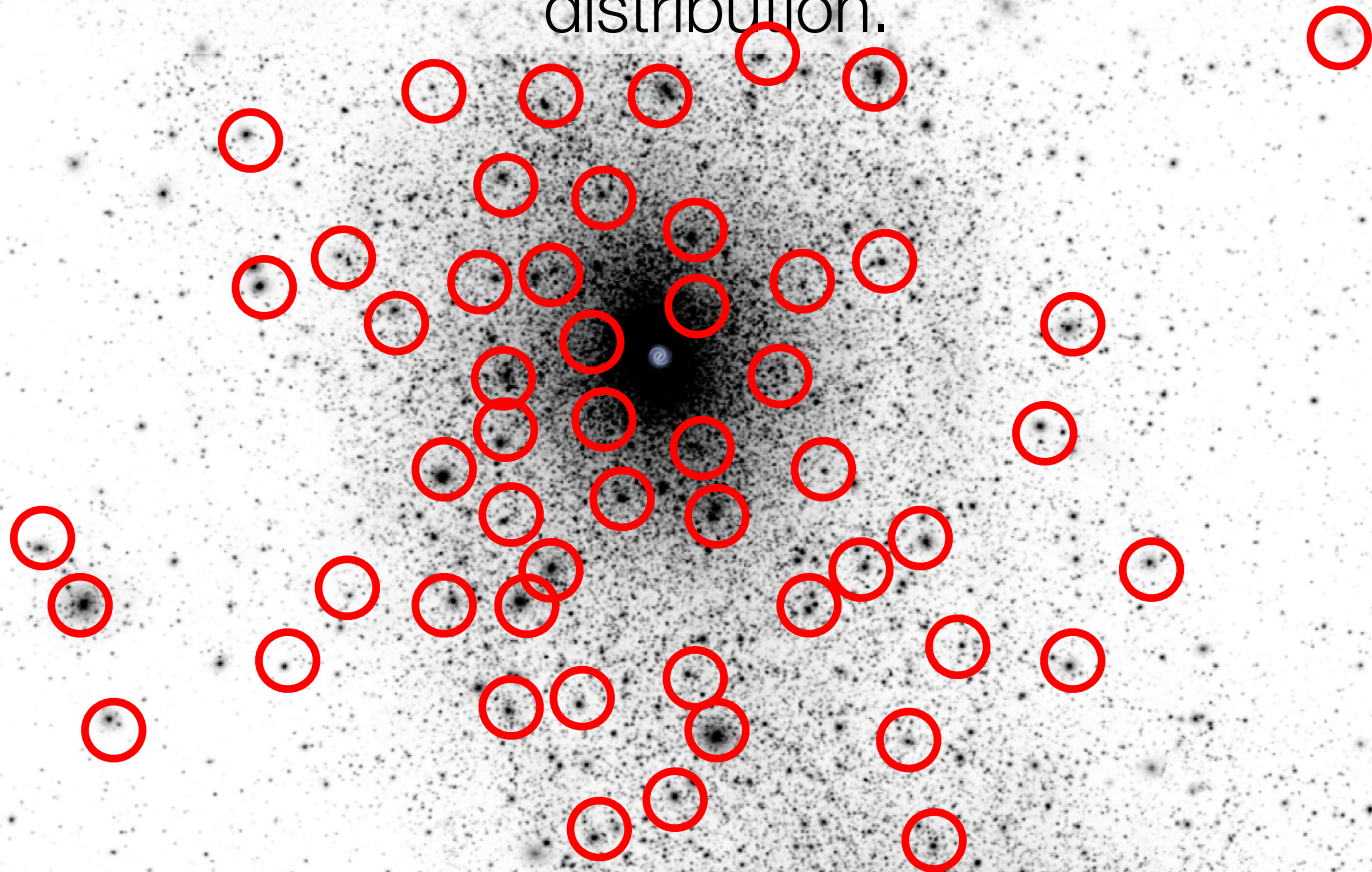


Our Milky Way's disk is believed to be embedded within a large distribution of dark matter



Fact

Our Milky Way hosts *many* satellite galaxies whose motions are dictated by the Milky Way's dark matter distribution.



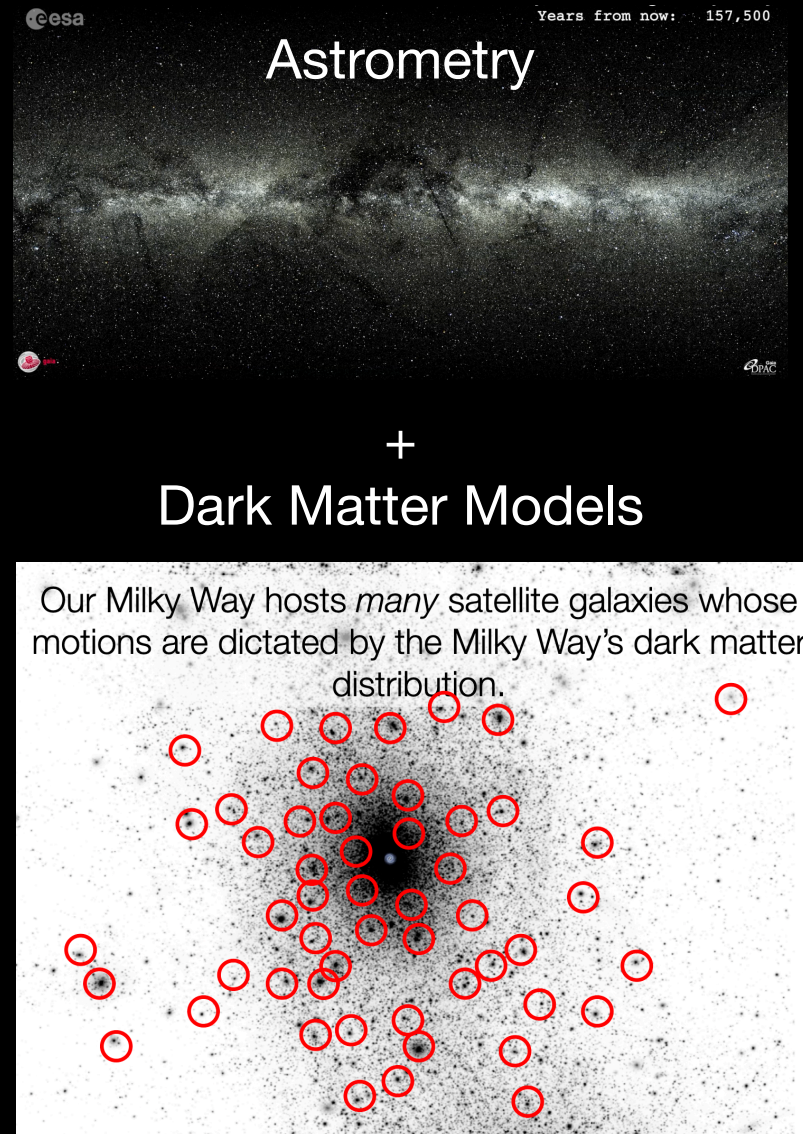
Fact

Astronomers are trying to use the
observed motions of
satellite galaxies orbiting the
Milky Way to understand the

distribution and total amount of
dark matter

in and around our Galaxy

Importance of Fact



The most massive satellite, the LMC, is on its first infall towards our Milky Way

Credit: Gaia

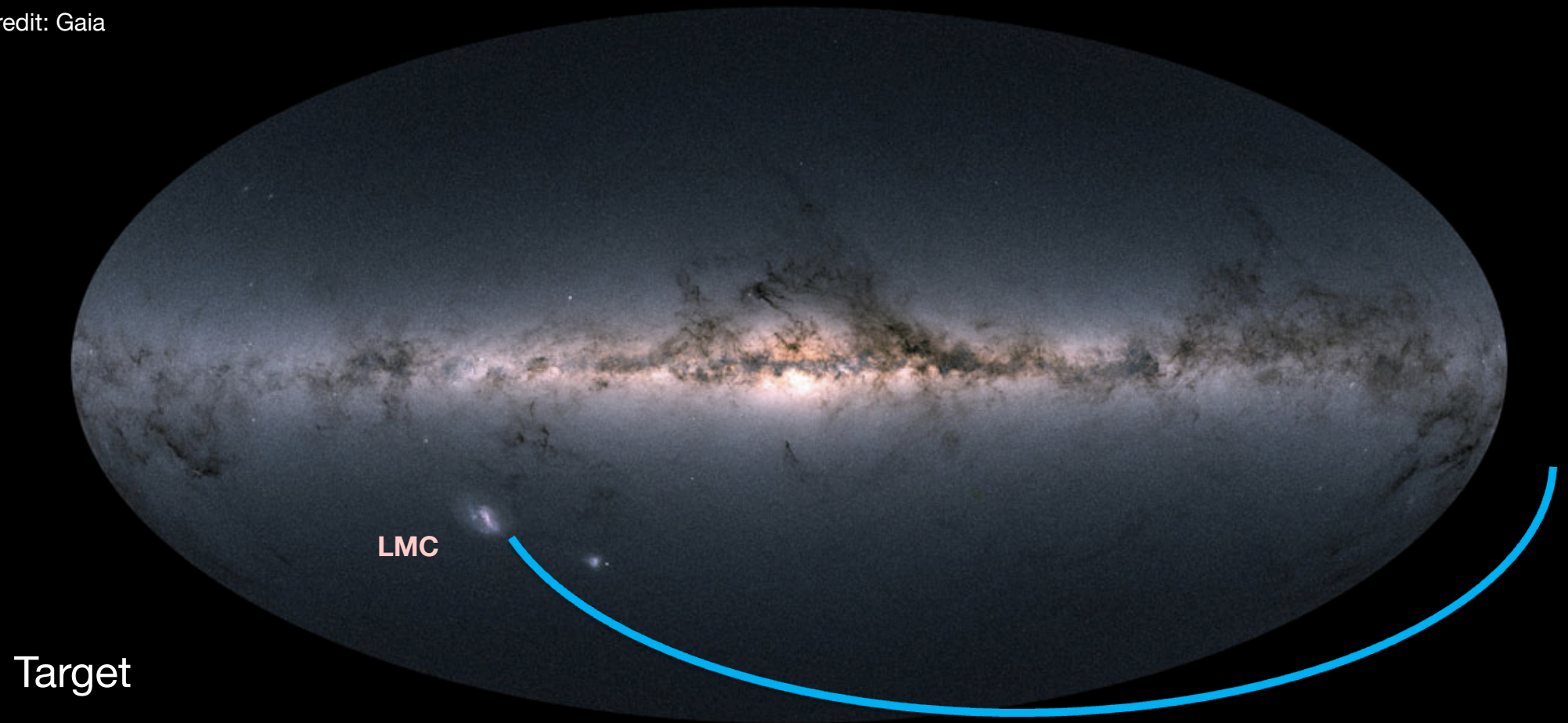


Image Credit: Carlos Milovic F.

African ethnoastronomy

Sotho: The tracks of celestial animals

Australian Indigenous Astronomy

Lower Murray: *Prolggi* “Cranes”

South America:

Mapuche: *Ruganko* (two wells)

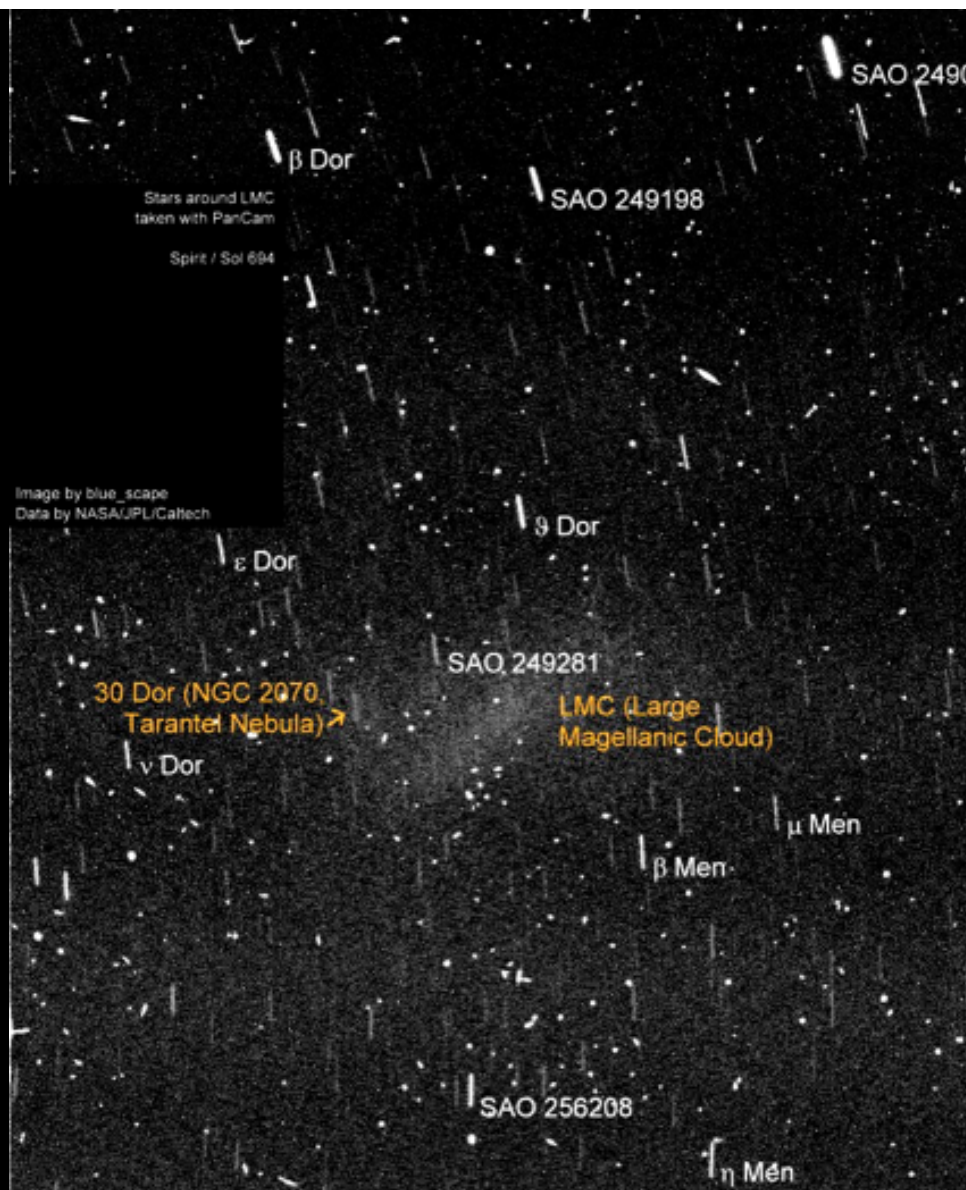
*See Article in APS
by Mia de los Reyes*

The LMC: A “massive” satellite galaxy

$V_{max} \sim 90 \text{ km/s}$, $M_{star} \sim 3e9 M_{\odot}$

Target [Side Slide for ref]

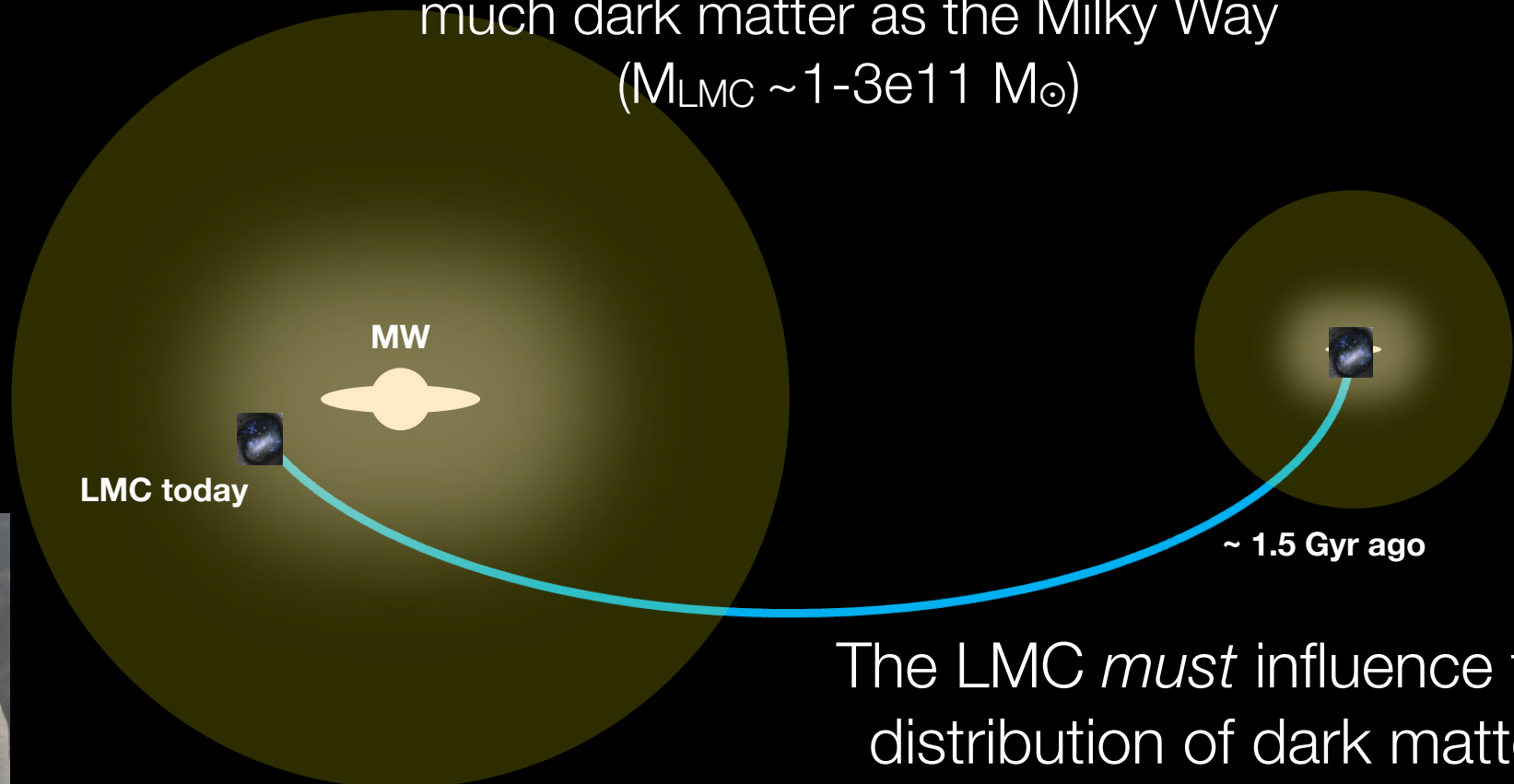
Image Credit: NASA



Target [Side Slide
for ref]

Problem

Dark matter theory predicts the LMC should have $\sim 1/10$ - $1/3$ as much dark matter as the Milky Way
($M_{\text{LMC}} \sim 1\text{-}3 \times 10^{11} M_{\odot}$)



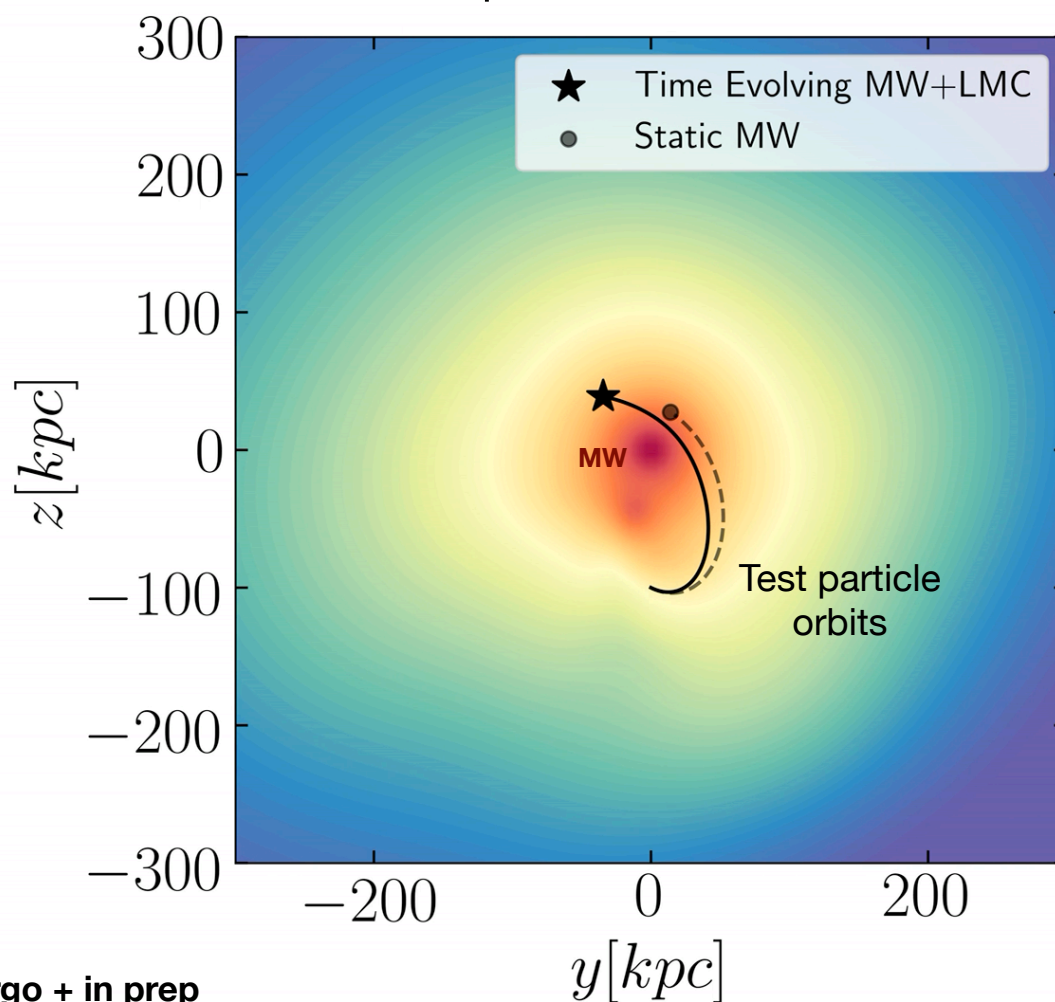
The LMC *must* influence the distribution of dark matter around our Milky Way and the orbits of all of its satellites



Speeds from the Hubble Space Telescope
(Era of Precision Astrometry)

Nitya Kallivayalil +2006, 2013
Besla+2007

N-Body simulations that account for the time evolution of the Milky Way+LMC dark matter distribution will provide a new tool for orbit integration.



Problem: Ignoring the LMC will result in incorrect orbit calculations.

The era of precision Astrometry is here, and yet we have no strategies to address the impact of the LMC.



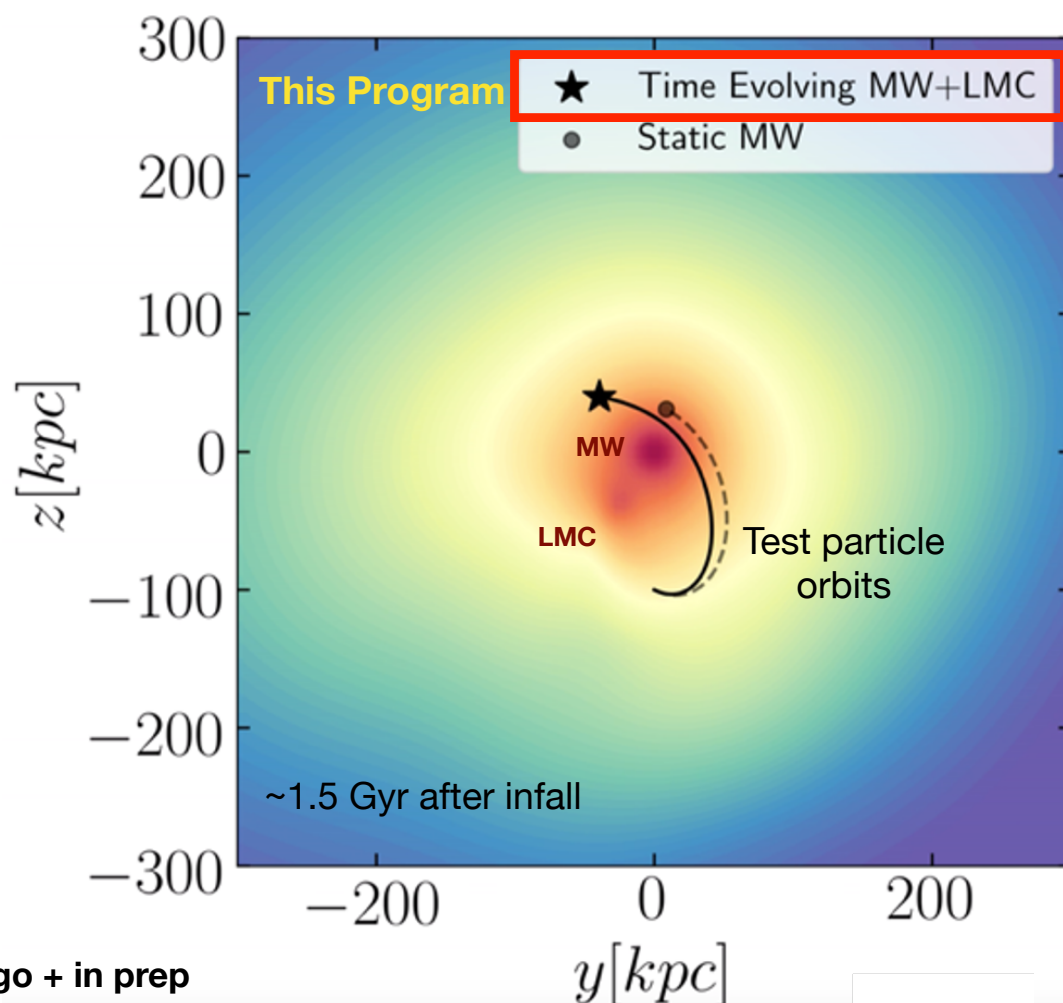
Nico Garavito-Camargo + in prep

N-Body simulations that account for the time evolution of the Milky Way+LMC dark matter distribution will provide a new tool for orbit integration.

Solution:
FIGURE
“This
proposal”



Nico Garavito-Camargo + in prep



Problem: Ignoring the LMC will result in incorrect orbit calculations.

The era of precision Astrometry is here, and yet we have no strategies to address the impact of the LMC.

Importance
of Soln &
Urgency

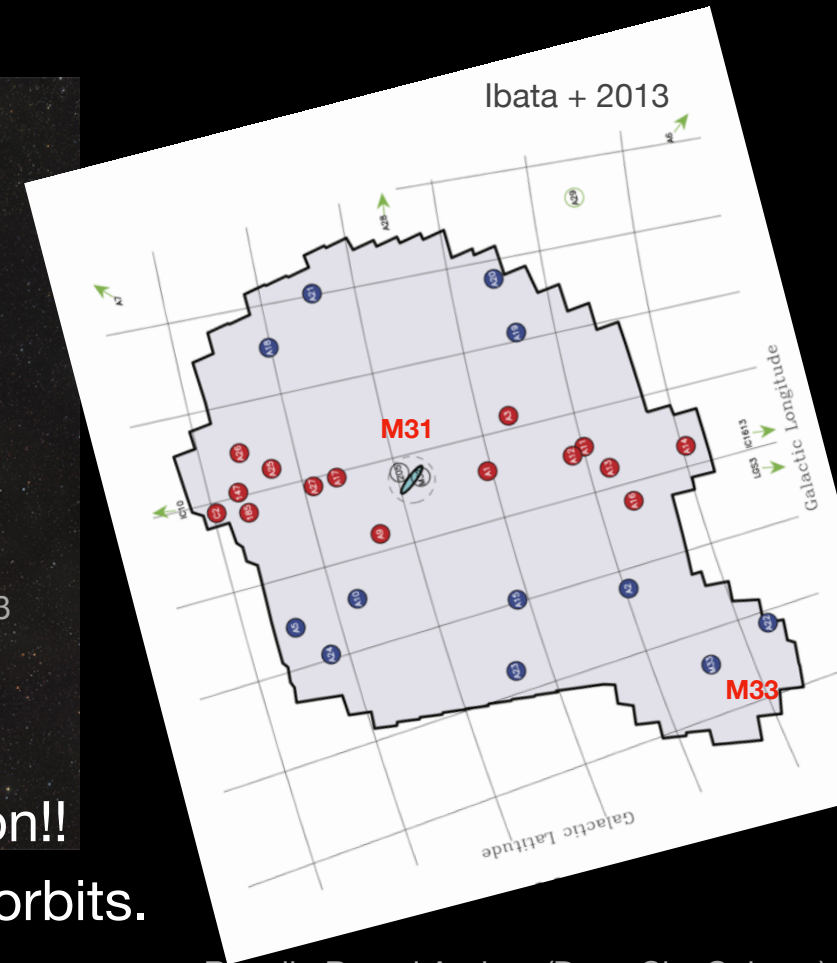
Broader
Implications:

M31 is the next frontier

M31 hosts a massive satellite (M33)

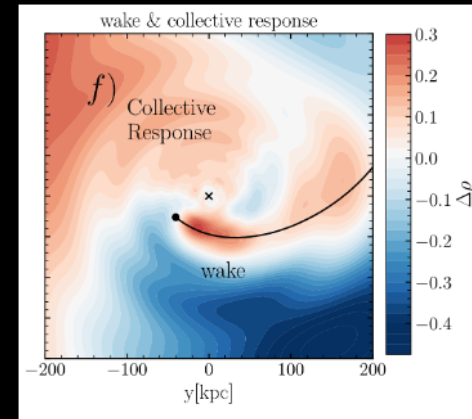
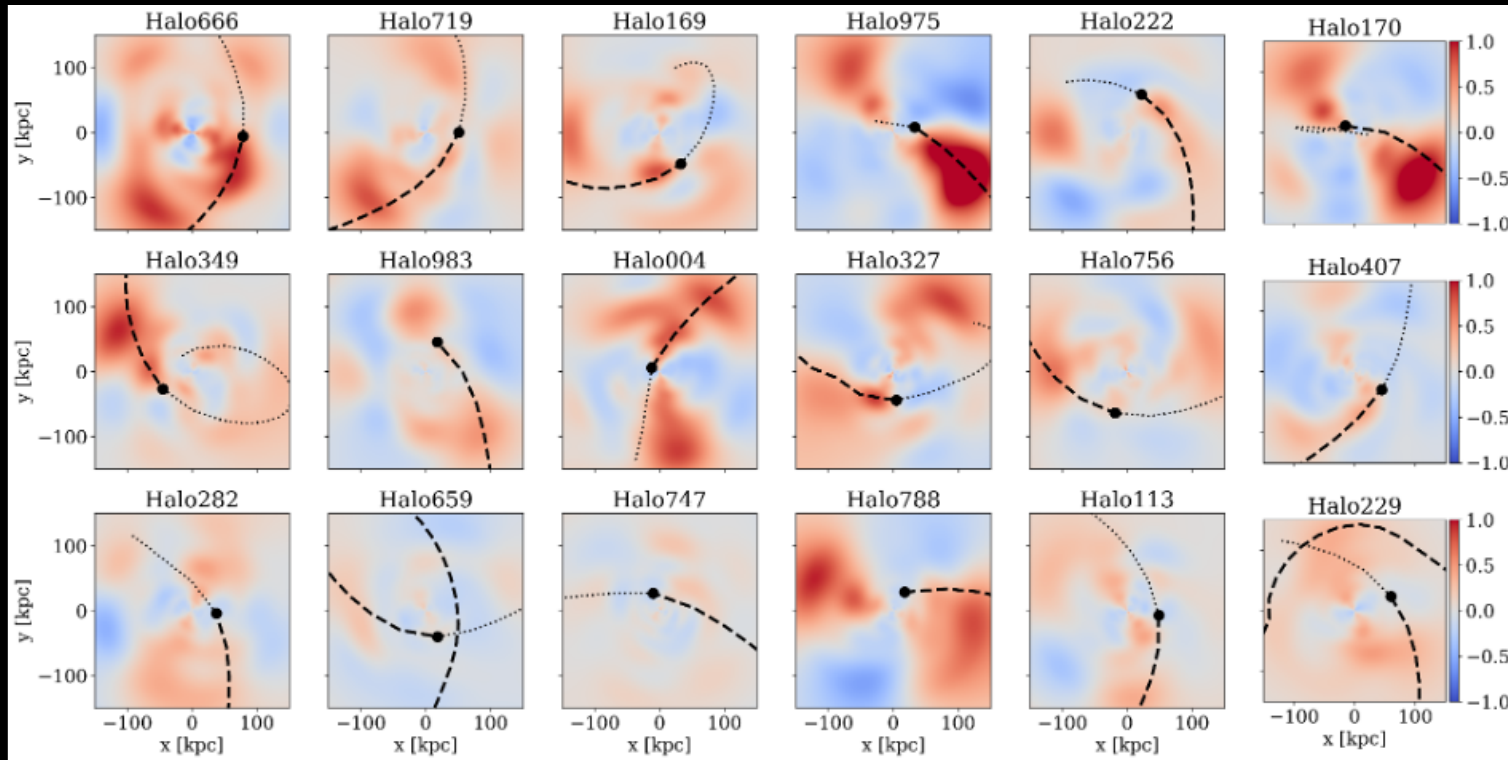


JWST astrometry for M31 satellites is coming soon!!
Time-evolving potentials are needed to integrate orbits.



Rogelio Bernal Andreo (Deep Sky Colours)

Cosmological Simulations show that all massive Satellites *change* the distribution of dark matter around their hosts



LMC Wake in the MW



Wakes in the MWest Cosmological simulations

Elise Darragh-Ford, Garavito-Camargo +2025

Alternative Broader Implications

Answering Questions

1) **Repeat back what the person said:** buys you time and you can make sure you are answering the correct question.

2) **Add context if needed:** Sometimes you need to give some context to the audience about the question - especially if there are students in the audience.

- **If you can't answer the question:** (you froze, you don't actually know, you can't remember)

Answering Questions

1) **Repeat back what the person said:** buys you time and you can make sure you are answering the correct question.

2) **Add context if needed:** Sometimes you need to give some context to the audience about the question - especially if there are students in the audience.

- **If you can't answer the question:** (you froze, you don't actually know, you can't remember)

First do 1 and 2.

THEN: That's a great question

THEN:

- Let's talk more after.
- But I don't know the answer to that off the top of my head, let me get back to you.

Types of Hostile Questions

Google AI :

1. Loaded Questions : (Assume Failure)

"Why did you **ignore** [specific paper] that completely **contradicts** your findings?" (Framed to accuse, not to explore findings)

"Are you still convinced this technique works despite the known **failures** in your pilot study?" (Invalidates the premise)

2. Aggressive/Dismissive Assertions

"This is just a rehash of [Name]'s work from 1990; what have you actually added?"

3. Misleading Comparisons

"Why should we trust your data when Dr. X found the opposite?" (Focusing on conflict rather than methodological differences).

4. Personal/Ad Hominem Attacks

"As someone not trained in [field], how can you justify these conclusions?". (Dismissing your expertise - particularly for students)

Don't Be a Conference Troll (the Guardian): <https://www.theguardian.com/higher-education-network/2015/nov/11/dont-be-a-conference-troll-a-guide-to-asking-good-questions>

Dealing with Hostile Questions

How have you dealt with Hostile Questions?

Dealing with Hostile Questions

In theory, the session chair should help..... But they won't, unless it drags on.
And even then, it depends on power dynamics.

FIRST :

- Assume it isn't hostile (doesn't mean it isn't). Most people are awkward and don't know how to ask questions.
- Switch the narrative in your head - opportunity to understand the counterpoints to your work so you can do better science.

NEXT:

- Rephrase the question to be a way that you can engage with. (Neutralize)
- Complement the question. Really glad you brought up this important point.
- If the question is falsely loaded, you can say so ("Respectfully disagree"). That helps the community in the room to understand the landscape and make an informed conclusion.

THEN

- In all cases, defer the conversation to later. Firmly. "This is worth a longer conversation - let's discuss after the talk."
- Then immediately launch into your next point and don't wait for a reply.